CE205

Lecture 8 Activities

Problem 1: What is the query-document match score that the Jaccard coefficient computes for each of the two documents below?

Query: march of dimes

Document 1: caesar died in march

Document 2: the long march

Jaccard(Q,D1)=?

Jaccard(Q,D2)=?

Problem 2: The tf-idf weights for the terms in various documents are given as follows:

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	5.25	3.18	0	0	0	0.35
Brutus	1.21	6.1	0	1	0	0
Caesar	8.59	2.54	0	1.51	0.25	0
Calpurnia	0	1.54	0	0	0	0
Cleopatra	2.85	0	0	0	0	0
mercy	1.51	0	1.9	0.12	5.25	0.88
worser	1.37	0	0.11	4.15	0.25	1.95

What will be the score of each document for the query ***'What does Antony say after Caesar dies?'***? What will be the order of the retrieved documents?

$$\text{Score}(q,d) = \sum_{t \in q \cap d} \text{tf.idf}_{t,d}$$

Problem 3: The bag-of-words model for the two documents and query is given as follows:

	Dog (T_1)	Man (T_2)	Bite (T_3)
D_1	2	3	5
D_2	3	7	1
Q	0	0	2

(Documents and queries are represented in the form of vectors in the figure below)

Compute the similarity score of the given query with each document using

(a) Inner product
(b) Cosine similarity

